

Complete the following steps for $f(x) = x^2 + 1$, interval $[0, 2]$, and $n = 4$.

1. Sketch the graph of the function on the given interval and illustrate the *left Riemann sums* using n subintervals.

2. Write an expression that gives the approximate value of the left Riemann if n is very large (you do not have to compute the value).

Suppose that $\int_0^3 f(x) dx = 2$ and $\int_3^6 f(x) dx = -5$, and $\int_3^6 g(x) dx = 1$. Evaluate the following integrals.

3. $\int_3^0 f(x) dx$

4. $\int_3^6 (2f(x) - g(x)) dx$

5. $\int_0^6 f(x) dx$

Write an expression which gives the average value of $f(z)$ on the given interval (you do not need to compute the value).

6. $f(z) = \sin(2z)e^{1-\cos(2\pi)}$ on the interval $[-\pi, \pi]$

The velocity of an object is, $\frac{ds}{dt} = -20t$ with initial position $s(0) = 100$. Use this information to solve the following two problems.

7. Find the position function $s(t)$.

8. Find the displacement of the object from $t = 1$ to $t = 2$.

Evaluate the area of the region described.

9. Write an integral expression which gives the area of the shaded region in the figure.

10. Find the area of the region bounded by the graphs of $f(x) = x^2 + 2$ and $g(x) = x - 2$ from $x = -1$ to $x = 1$.

Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved about the x -axis.

11. $y = \sqrt{x+1}$ from $x = 0$ to $x = 3$.

Find the length of the curve for the given function and interval.

12. $y = \frac{2}{3}x^{3/2} + 9$ from $x = 0$ to $x = 3$.

Evaluate the following integrals.

13. $\int_{-1}^1 6x^5 - x + 5 \, dx$

14. $\int 20(5x + 1)^{-4} \, dx$

15. $\int \frac{t^5 - 5t + 2}{t^2} \, dt$

16. $\int \left(\frac{2}{x} - 4e^{2x} \right) dx$

Evaluate the following integrals by using integration by parts.

17. $\int x e^{-x} dx$

18. $\int 5x \sin x dx$

Evaluate the integrals.

19. $\int \cos^2 x dx$

20. $\int \sin^3 x \cos^2 x \, dx$

21. $\int \frac{6}{x^2 + 2x - 8} \, dx$

22. $\int \frac{dx}{\sqrt{9 - x^2}}$

23. $\int_0^2 \sqrt{4-x^2} \, dx$

Evaluate the derivative

24. $\frac{d}{dx} \frac{2}{3} e^{3x}$

25. $\frac{d}{dx} (\ln(4x+1) + 5)$